STATISTICS

Analysis of Variance (ANOVA)

Birol Aslanyürek

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The Means of More Than Two Populations

We apply the t-test or z-test to compare the means of two populations. When we compare the means of more than two populations, one may try one of these tests multiple times. In this case, however,

- You should apply $C\binom{M}{2}$ individual hypothesis testing.
- The probability of type I error, α, increases as the population number increases.

For example, when $M = 5$ populations and $\alpha = 0.05$,

$C\binom{5}{2} = 10$ hypothesis testing are applied and confidence level is:

$$(1-\alpha\,)^{10} = (0.95)^{10} \cong 0.60 \rightarrow \alpha \cong 0.40$$

Analysis of Variance (ANOVA)

Statistics

Analysis of Variance (ANOVA)

Analysis **o**f **va**riance (ANOVA) is a more appropriate method when comparing the means of more than two populations.

Assumptions of ANOVA:

1. The populations have a normal distribution.
2. The homogeneity of variances (The populations have equal variances.)
3. The populations are independent. Samples are also independent and randomly selected.

Types of ANOVA

Consider the following examples to which ANOVA can be applied:

- The expected life of tires from 5 different brands: One factor (the brands) affects the results.
- The mean grades in Statistics, Mathematics and English: One factor (the type of course) affects the results.
- The mean grades in Statistics, Mathematics, and English for male and female students: Two factors (the type of course and the gender) affect the results.
- What is the relationship between health insurance payments of a person with the career type and marital status: Two factors (the career type and marital status) affect the health insurance payments.

ANOVA

One-way ANOVA
for the problems including one factor

Two-way ANOVA
for the problems including two factors

Analysis of Variance (ANOVA)

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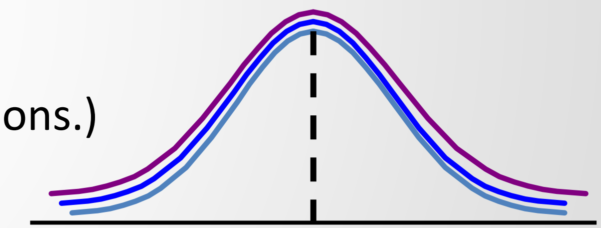
One-Way ANOVA

Consider M populations having normal distribution with equal variances and means $\mu_1, \mu_2, \mu_3, \ldots, \mu_M$.

Hypotheses:

$H_0 \colon \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_M$

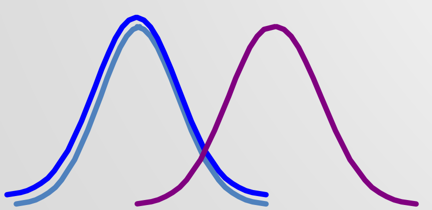
(The factor has no effect on the means of the populations.)



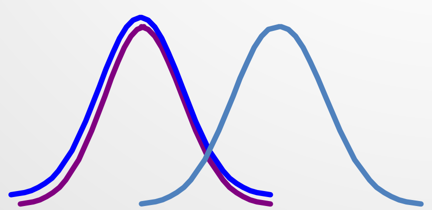
$\mu_1 = \mu_2 = \mu_3$

H_a: At least one of the population mean is different

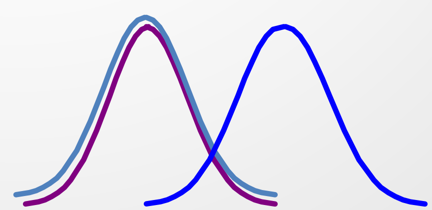
(The factor affects the means of the populations.)



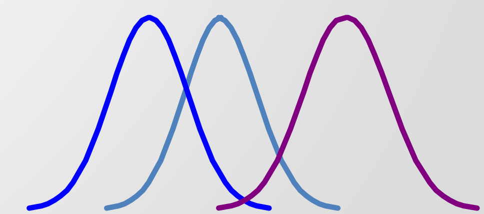
$\mu_1 = \mu_2 \neq \mu_3$



$\mu_1 = \mu_3 \neq \mu_2$



$\mu_2 = \mu_3 \neq \mu_1$



$\mu_1 \neq \mu_2 \neq \mu_3$

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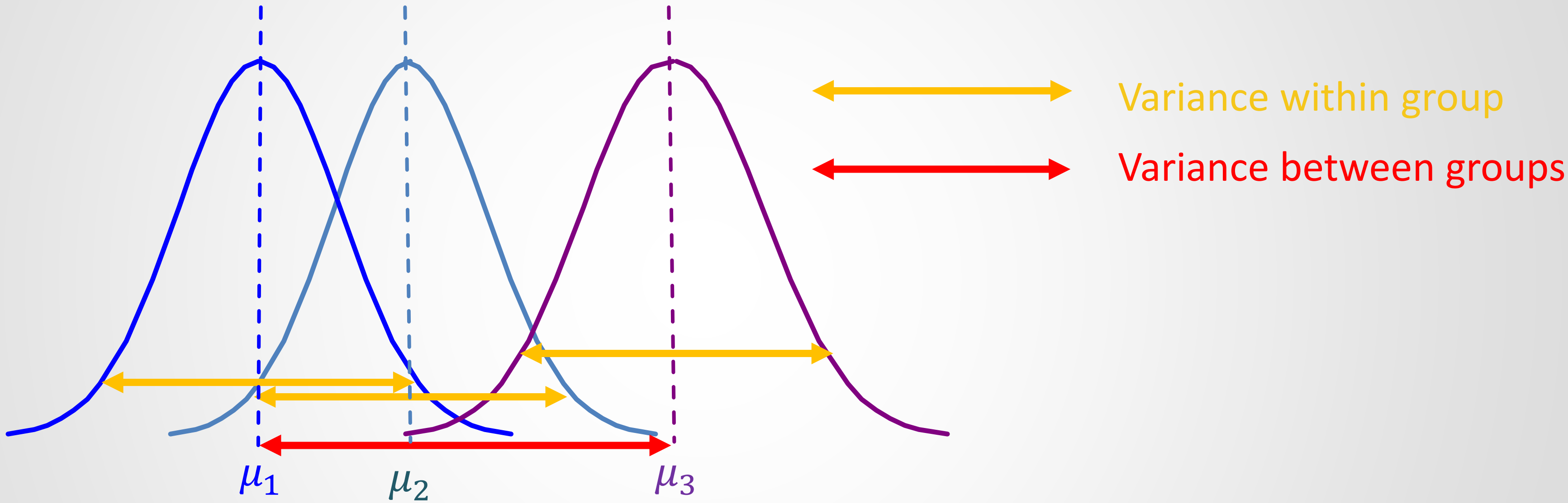

Test Statistic:

$$F_0 = \frac{Variance\ Between\ Groups}{Variance\ Within\ Group}$$

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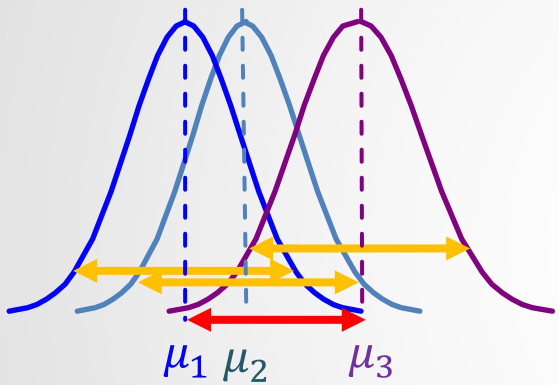
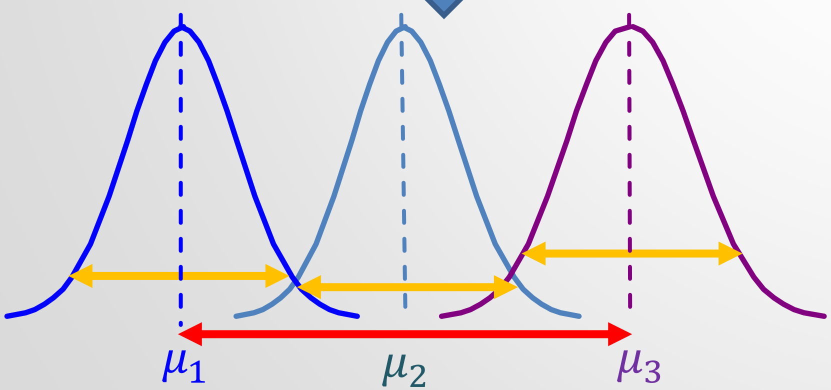
As variance between groups increases, test statistic F_0 increases.



$$F_0 = \frac{Variance\ Between\ Groups}{Variance\ Within\ Group}$$

The means diverge from each other.



implies that H_a is correct

H_a: At least one of the population mean is different
(The factor affects the means of the populations.)

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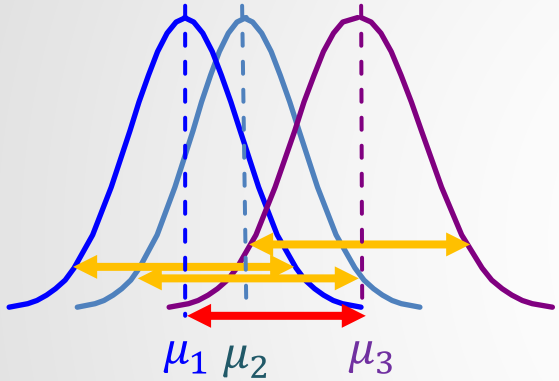
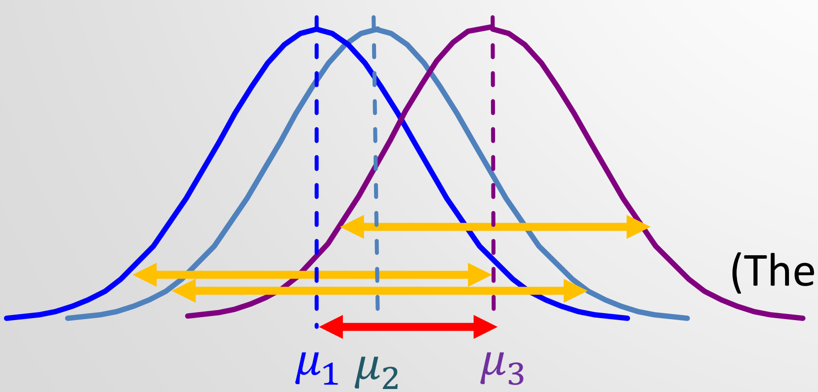
As variance within group increases, test statistic F_0 decreases.



$$F_0 = \frac{Variance\ Between\ Groups}{Variance\ Within\ Group}$$

The intersection area of the populations is increased.



implies that H_0 is correct

$$H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_M$$

(The factor has no effect on the means of the populations.)

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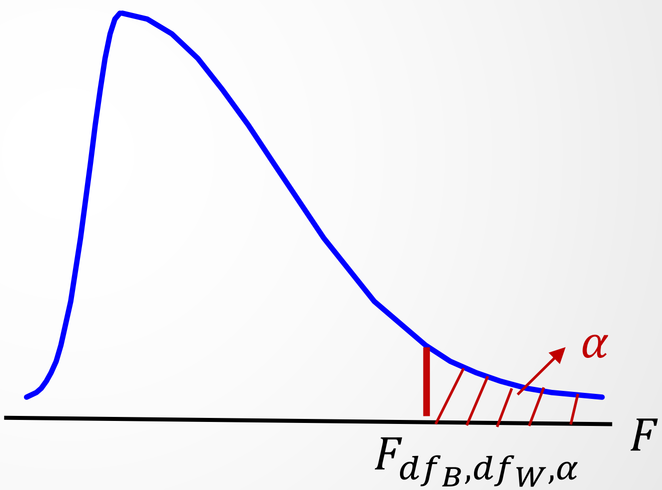
Decision Rule:

If

$$F_0 \leq F_{df_B, df_W, \alpha}$$

accept H_0.

Otherwise, reject H_0.



$$\mathrm{P}\left(F_0 \leq F_{df_B, df_W, \alpha}\right) = 1 - \alpha$$

Analysis of Variance (ANOVA)

Statistics

Computation of F_0

The measurements of M samples are given below.

	Factor 1				
	Group 1	Group 2	…	Group k	
	x_{11}	x_{21}	…	x_{k1}	
	x_{12}	x_{22}	…	x_{k2}	
	…	…	…	…	
	…	x_{2n_2}	…	…	
	x_{1n_1}		…	x_{kn_k}	TOTAL
sample size	n_1	n_2	…	n_k	$n = \sum_{i=1}^{k} n_i$
sum of sample	$T_1 = \sum_{j=1}^{n_1} x_{1j}$	$T_2 = \sum_{j=1}^{n_2} x_{2j}$	…	$T_k = \sum_{j=1}^{n_k} x_{Mj}$	T= $\sum_{i=1k} T_i$
sample mean	$\bar{x}_1 = {T_1}/{n_1}$	$\bar{x}_2 = {T_2}/{n_2}$	…	$\bar{x}_k = {T_k}/{n_k}$	$\bar{x} = {T}/{n}$

Analysis of Variance (ANOVA)

Statistics

Computation of F_0

$$F_0 = \frac{Variance\ Between\ Groups}{Variance\ Within\ Group} = \frac{SS_B/df_B}{SS_W/df_W}$$

The sum of squares between groups:

$$SS_B = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2 = n_1(\bar{x}_1 - \bar{x})^2 + n_2(\bar{x}_2 - \bar{x})^2 + \cdots + n_k(\bar{x}_k - \bar{x})^2$$

The degree of freedom between groups:

$$df_B = k - 1$$

The sum of squares within group:

$$SS_W = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij} - \bar{x}_i\right)^2 = \sum_{i=1}^{k}(n_i - 1)s_i^2 \text{ where } s_i^2 = \frac{1}{n_i - 1}\sum_{j=1}^{n_i}\left(x_{ij} - \bar{x}_i\right)^2$$

The degree of freedom between groups:

$$df_W = n - k$$

the variance of the sample (group) i

Analysis of Variance (ANOVA)

Statistics

Computation of F_0

The total sum of squares:

$$SS_T = SS_B + SS_W$$

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij}-\bar{x}\right)^2 = \sum_{i=1}^{k} n_i(\bar{x}_i-\bar{x})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij}-\bar{x}_i\right)^2$$

The total degree of freedom:

$$df_T = df_B + df_W$$

$$n-1 = k-1+n-k$$

Analysis of Variance (ANOVA)

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Proof for $SS_T = SS_B + SS_W$

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij}-\bar{x}\right)^2 = \sum_{i=1}^{k} n_i(\bar{x}_i-\bar{\boldsymbol{x}})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij}-\bar{x}_i\right)^2$$

$$SS_B + SS_W = \sum_{i=1}^{k} n_i(\bar{x}_i-\bar{\boldsymbol{x}})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij}-x_i\right)^2$$

$$= \cancel{\sum_{i=1}^{k} n_i(\bar{x}_i)^2} - 2\bar{\boldsymbol{x}}\sum_{i=1}^{k} n_i\bar{x}_i + (\bar{\boldsymbol{x}})^2\sum_{i=1}^{k} n_i + \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij}\right)^2 - 2\cancel{\sum_{i=1}^{k}\bar{x}_i\overbrace{\sum_{j=1}^{n_i} x_{ij}}^{n_i\,\bar{x}_i}} + \cancel{\sum_{i=1}^{k}(\bar{x}_i)^2\overbrace{\sum_{j=1}^{n_i} 1}^{n_i}}$$

$$\bar{X}_i = \frac{\sum_j^{n_i} X_{ij}}{n_i} \qquad n_i = \sum_{j=1}^{n_i} 1$$

$$= -2\bar{\boldsymbol{x}}\sum_{i=1}^{k}\sum_{j} x_{ij} + (\bar{\boldsymbol{x}})^2\sum_{i=1}^{k}\sum_{j=1}^{n_i} 1 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij}\right)^2$$

$$= \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left[\left(x_{ij}\right)^2 - 2\bar{\boldsymbol{x}}x_{ij} + (\bar{\boldsymbol{x}})^2\right] = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij}-\bar{\boldsymbol{x}}\right)^2 = SS_T$$

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Mean Square (variance)

The mean squares between groups:

$$MS_B = \frac{SS_B}{df_B} = \frac{SS_B}{k-1}$$

$$F_{df_B, df_W, \alpha} = F_{k-1, n-k, \alpha}$$

The mean squares within groups:

$$MS_W = \frac{SS_W}{df_W} = \underbrace{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2 + \cdots + (n_k-1)s_k^2}{(n_1-1) + (n_2-1) + \cdots + (n_k-1)}}_{\text{the weighted mean of the variances } s_i^2} = \frac{SS_W}{n-\mathrm{k}}$$

Analysis of Variance (ANOVA)

Statistics

One-Way ANOVA Table

Source of the variability	Sum of Squares (SS)	Degrees of freedom (df)	Mean Square (Variance)	F_0
Between Groups	SS_B	$k-1$	$MS_B = \frac{SS_B}{k-1}$	$\frac{MS_B}{MS_W}$
Within Groups (Residual-Error)	SS_W	$n-k$	$MS_W = \frac{SS_W}{n-k}$	
Total	SS_T	$n-1$		

M: the number of populations (groups)

n: the sum of sample sizes

The Total Variation

$$Total\ variance = Variance\ Between\ Groups + Variance\ Within\ Group$$

The variation due to the effect of the factor
+
The variation due to the random errors (chance factor)

The variation due to the random errors (chance factor)

$H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_M$
(The factor has no effect on the means of the populations.)

When H_0 is correct,

$$F_0 = \frac{Variance\ Between\ Groups}{Variance\ Within\ Group} \approx 1$$

Analysis of Variance (ANOVA)

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The Mathematical Model of One-Way ANOVA

The problem addressed in ANOVA can be modeled by the following formula:

$$X_{ij} = \mu_i + \varepsilon_{ij}$$

where X_{ij} is the j.member of sample i, μ_i is the mean of the population i and ε_{ij} is the random error in the j.member of sample i. Then, if the mean of all populations μ (without the effect of the factor) is subtracted from both sides,

$$X_{ij} - \mu = \mu_i + \varepsilon_{ij} - \mu$$

$$X_{ij} = \mu + (\mu_i - \mu\ ) + \varepsilon_{ij}$$

If we substitute $\alpha_i = \mu_i - \mu$, then

$$X_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

Here, α_i is the effect of the factor for the group i. Thus, hypotheses can be expressed as:

$$H_0: \alpha_1 = \alpha_2 = \cdots = \alpha_M = 0$$

H_a: at least one of $\alpha_i \neq 0$

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ANOVA vs MANOVA

The mathematical model of ANOVA is:

$$X_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

This formula contains a single dependent variable, X_{ij}.

When the formula contains a multivariate dependent variable, multivariate analysis of variance (MANOVA) should be used.

Hypothesis Testing

Statistics

Example 1: The out-of-service times of 23 photocopiers of three different brands in a company in the last year were determined in minutes, and the following data set was obtained. Can we say that there is a difference between the three machines in terms of the mean out-of-service times? Also, test the hypothesis by computing the p-value.

Machine A	Machine B	Machine C
64	45	75
72	62	85
81	73	83
92	58	78
80	50	96
73	68	88
	48	83
	52	84
	66	

(Assume that populations have normal distributions with equal variances and $\alpha = 0.01$)

Hypothesis Testing

Statistics

Machine A	Machine B	Machine C	
64	45	75	
72	62	85	
81	73	83	
92	58	78	
80	50	96	
73	68	88	
	48	83	
	52	84	
	66		
$n_1 = 6$	$n_2 = 9$	$n_3 = 8$	$n = 23$
$T_1 = 462$	$T_2 = 522$	$T_3 = 672$	T= 1656
$\bar{x}_1 = 77$	$\bar{x}_2 = 58$	$\bar{x}_3 = 84$	$\bar{x} = 72$

H_0: $\mu_1 = \mu_2 = \mu_3$ (The machine brands have no effect on the out-of-service times.)

H_a: $\mu_i \neq \mu_j$ for any $i \neq j$
(The machine brands affect the out-of-service times.)

$$SS_B = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2$$
$$= 6(77 - 72)^2 + 9(58 - 72)^2 + 8(84 - 72)^2$$
$$= 3066$$

$$df_B = M - 1 = 3 - 1 = 2$$

Hypothesis Testing

Statistics

Machine A	Machine B	Machine C	
64	45	75	
72	62	85	
81	73	83	
92	58	78	
80	50	96	
73	68	88	
	48	83	
	52	84	
	66		
$n_1 = 6$	$n_2 = 9$	$n_3 = 8$	$n = 23$
$T_1 = 462$	$T_2 = 522$	$T_3 = 672$	T= 1656
$\bar{x}_1 = 77$	$\bar{x}_2 = 58$	$\bar{x}_3 = 84$	$\bar{x} = 72$

$$SS_W = \sum_{i=1}^{k} \sum_{j=1}^{n_i} \left(x_{ij} - \bar{x}_i\right)^2$$

$$= (64-77)^2 + (72-77)^2 + (81-77)^2 + (92-77)^2 + (80-77)^2 + (73-77)^2 + (45-58)^2 + (62-58)^2 + (73-58)^2 + (58-58)^2 + (50-58)^2 + (68-58)^2 + (48-58)^2 + (52-58)^2 + (66-58)^2 + (75-84)^2 + (85-84)^2 + (83-84)^2 + (78-84)^2 + (96-84)^2 + (88-84)^2 + (83-84)^2 + (84-84)^2 = 1514$$

$$df_W = n - M = 23 - 3 = 20$$

Analysis of Variance (ANOVA)

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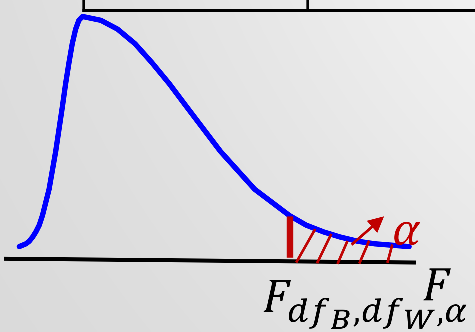
Example 1:

One-Way ANOVA Table

	Sum of Squares (SS)	Degrees of freedom (df)	Mean Square (Variance)	F_0
Between Groups	3066	$3-1=2$	$\frac{3066}{2}=1533$	$F_0=\frac{1533}{75.7}\cong \mathbf{20.25}$
Within Groups	1514	$23-3=20$	$\frac{1514}{20}=75.7$	
Total	4580	$23-1=22$		



$F_{2,20,0.01}=5.85$

Since $F_0=20.25>F_{df_B,df_W,\alpha}=5.85$,
we reject H_0.

We have very significant proof that there is a difference between the three machines in terms of the mean out-of-service times.

Analysis of Variance (ANOVA)

Statistics

Example 1:

F Table
$F_{2,20,0.01} = 5.85$
$F_{2,20,0.005} = 6.99$

$$F_{2,20,p\ value} = F_0 = 20.25 \Rightarrow p\ value \approx 0$$

Since $p\ value < \alpha$, we reject H_0.

We have very significant proof that there is a difference between the three machines in terms of the mean out-of-service times.